Taewoon Kang

Integrated M.S. and Ph.D. Student
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EDUCATION

Mar. 2023 ~ Present	**Korea University** Department of Computer Science and Engineering *Advisor: Gunjae Koo* *Ph.D. Student* GPA: 4.35 / 4.5	Seoul, Korea
Mar. 2019 ~ Aug. 2022	**Sangmyung University** Department of System Semiconductor Engineering Thesis: An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano *Advisor: Yongwoo Kim* *B.S in System Semiconductor Engineering* GPA: 4.28 / 4.5	Cheonan, Korea

RESEARCH INTERESTS

- AI Accelerator
- HW/SW Co-Design SoC
- Processing for AI (NPU)
- Computer architecture
- Processing-in memory

RESEARCH EXPERIENCES

- Visiting Scholar at Ming Hsieh Department of Electrical and Computer Engineering, University of Southern California , United States (Jan. 2026 ~ Present) / Conducting collaborative research as a Visiting Scholar under the supervision of Prof. Annavaram
- Intern at SK Telecom, Korea (Jul. 2023 ~ Aug. 2023) / Develop spam detection methods using AI

PUBLICATIONS (INTERNATIONAL CONFERENCE/JOURNA

1. Taewoon Kang,Namhun Kim,Jihun Lee,Gunjae Koo, "SoftmaxPIM: An HBM-Based PIM Architecture for Accelerating GeMV–Softmax Execution Pipeline", *IEEE COMPUTER*

ARCHITECTURE LETTERS, (2026)

2. Taewoon Kang,Namhun Kim,Geonwoo Choi,Taeweon Suh,Gunjae Koo, "SparsePIM+: Accelerating SpMV on HBM-Based PIM via Logic-Die Accumulators with Opportunistic TSV Utilization", *JOURNAL OF SYSTEMS ARCHITECTURE*, (2026)
3. Sunchae Kim,Taewoon Kang,Sangwon Shin,Taeweon Suh,Yibin Yang,Gunjae Koo, "SumcheckPIM: An Efficient HBM-Based PIM Architecture for Linear Complexity Zero Knowledge Proofs", *ACM INTERNATIONAL CONFERENCE ON SUPERCOMPUTING*, (2026)
4. Taewoon Kang,Geonwoo Choi,Taeweon Suh,Gunjae Koo, "SparsePIM: An Efficient HBM-Based PIM Architecture for Sparse Matrix-Vector Multiplications", *ICS '25: PROCEEDINGS OF THE 39TH ACM INTERNATIONAL CONFERENCE ON SUPERCOMPUTING*, (2025)

PUBLICATIONS (DOMESTIC CONFERENCE/JOURNAL)

1. Seongpil Yang, Taewoon Kang, "Performance Analysis of a Per-Row Activation Counting Mechanism in DRAM", The Institute of Electronics and Information Engineers, 2025 summer conference, (2025)
2. Sang-un Jo, Tae-woon Kang, Yongwoo Kim, "A Design and Implementation of MP3 Player using ARM Cortex-M0 DesignStart", The Korean Institute of Electrical Engineers Workshop, Yeosu, Korea (Jul. 2022)
3. Sang-un Jo, Tae-Woon Kang, Yongwoo Kim, "A design and verification of ARM Cortex-M0 DesignStart MCU system using open source IP", ICS 2022, Jeonju, Korea (Apr. 2022)
4. Taewoon Kang, Yongwoo Kim, "An Implementation of CNN-based Real-Time Face Mask Detector using Jetson Nano ", JKAIS, (Vol. 23, 2022)

AWARDS AND HONORS

- The top prize of CISC Career day, Sangmyung University, Korea (May. 2022)
- Header (in one's dept.) Scholarship, Sangmyung University, Korea (Mar. 2022)
- Header (in one's dept.) Scholarship, Sangmyung University, Korea (Sep. 2021)
- "BONGSA" Scholarship (Leader of Department), Sangmyung University, Korea (Sep. 2021)
- Secondary Header (in one's dept.) Scholarship, Sangmyung University, Korea (Mar. 2021)
- Work Scholarship, Sangmyung University, Korea (Mar. 2021)
- "BONGSA" Scholarship (Leader of Department), Sangmyung University, Korea (Mar. 2021)
- Secondary Header (in one's dept.) Scholarship, Sangmyung University, Korea (Sep. 2020)

COURSE WORK

- Parallel processing, AI accelerator architecture
- Advanced computer architecture, Storage system architecture
- ARM Cortex-M series IP implementation
- Electrical circuit, Microelectronic circuit
- Semi-custom ASIC design with Synopsys design tool, VLSI design
- Digital logic circuit, Computer architecture and design, Design with Verilog, SoC design
- Semiconductor Physics, Semiconductor fabrication process
- Calculus, Engineering mathematics

Skills and Techniques

- C, C++, Python
- GEM5, DRAMsim3
- Chip design verification with Xilinx/Altera FPGA
- Python (Pytorch)/ C language (Atmel, Keil)
- Verilog 2001 / SystemVerilog
- ARM Cortex-M0/M3 implementation
- RISC-V Simple processor design
- Semi-custom ASIC Design with Synopsys EDA tool

Teaching Experience

- Teaching Assistant
  - Basic of software: Spring 2025
  - Digital logic design: Spring 2023, Fall 2023